

a

Dropout

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Dropout midpoint	1	0	1	2	3					
PCA	7	NR	6	2	4					
PcaGrid	10	NR	10	10	10					
PcaHubert	9	NR	9	8	8					
PCP	1	NR	1	9	9					
K's tau	2	NR	3	1	1					
Winsor	5	NR	7	4	3					
Quad	4	NR	5	3	2					
Ball	6	NR	2	5	5					
Shell	8	NR	8	7	7					
LR	3	NR	4	6	6					

c

Gene-subset mean shift

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Shifted-cell fraction	2%	4%	6%	8%	10%					
PCA	6	2	5	6	6					
PcaGrid	10	10	10	10	10					
PcaHubert	9	9	9	9	9					
PCP	1	1	1	2	1					
K's tau	4	7	4	5	3					
Winsor	8	8	7	7	7					
Quad	7	5	8	8	8					
Ball	3	3	2	3	4					
Shell	5	6	6	4	5					
LR	2	4	3	1	2					

b

Synthetic doublets

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Doublet fraction	2%	5%	8%	10%	15%					
PCA	2	2	3	2	5					
PcaGrid	10	10	10	10	10					
PcaHubert	9	9	9	8	9					
PCP	1	1	1	1	1					
K's tau	8	8	8	9	8					
Winsor	5	5	4	3	3					
Quad	6	6	5	6	4					
Ball	3	3	6	7	7					
Shell	4	4	2	4	2					
LR	7	7	7	5	6					

d

Gene-subset mean shift

	PCA	PcaGrid	PcaHubert	PCP	K's tau	Winsor	Quad	Ball	Shell	LR
Shift magnitude	1.5	2	3	4	5					
PCA	8	6	6	6	7					
PcaGrid	10	10	10	10	10					
PcaHubert	9	9	9	9	9					
PCP	4	1	1	1	1					
K's tau	2	4	5	7	6					
Winsor	7	8	7	5	4					
Quad	6	7	8	8	8					
Ball	3	3	3	4	5					
Shell	5	5	4	3	3					
LR	1	2	2	2	2					

